BILLY HERNAWAN

Quantitative Risk Scientist & PhD Candidate specializing in Probabilistic Risk Assessment, causal inference, and uncertainty quantification for safety-critical physical systems

217-706-2252 ■ hernawa2@tamu.edu ■ linkedin.com/in/hernawa2 ■ hernawa2.github.io

EDUCATION

Texas A&M University College Station, TX
PhD *in Civil Engineering with a focus on Bayesian Risk Modeling* Expected Dec 2026
Minor *in Applied Statistics*
Thesis Topic: "Bayesian Inference & Stochastic Modeling for Complex Hazard and Risk Assessment"
Relevant Coursework: Database & Computational Tools for Big Data, Advanced Statistical Machine Learning, Deep Learning, Deep Reinforcement Learning, Stochastic System & Design, Bayesian Modeling & Computation.

University of Illinois Urbana-Champaign *Champaign, IL*
BS & **MS** in Civil Engineering *May 2018 - Dec 2019*

RESEARCH EXPERIENCE

Stochastic Geomechanics Laboratory **College Station, TX**
Research Assistant June 2021 – Present

Causal Supply Chain Risk: Graphical Machine Learning for Causal Inference in U.S. Supply Chains

- Developed a probabilistic parameter learning algorithm (**Markov Chain Monte Carlo**) to train a large-scale graphical causal model (**Bayesian Network**) to simulate risk scenarios and aid decision-making strategy
- Constructed high-dimensional conditional probability tables using **Hamiltonian Monte Carlo (HMC)** sampling to quantify uncertainty and evaluate low-probability tail risks across complex dependencies.
- Constructed a **constrained Monte Carlo** as a validation tool for Bayesian Risk Network topology
- Formulated a cloud-based **end-to-end machine learning & data wrangling environment** to facilitate data science convergence among 6 independent working groups of different expertise in the US, Mexico and Canada.

Random-Field Offshore Landslide Risk Assessment

- Constructed rare-event spatial risk models and dynamic programming algorithms to optimize path routing under extreme environmental uncertainties.

Machine Learning-Based Optimization for Offshore Pipeline Routing

- Developed a Dynamic Programming to optimize pipeline routing on a landslide hazard map and benchmarked results with common shortest path optimization technique.
- Established a combined and independent framework linking engineering risk and geological analysis for offshore pipeline routing using probability operator.

University of Illinois Urbana-Champaign **Champaign, IL**
Research Assistant May 2018 – Dec 2019

BNSF Railway Track Instrumentation for Remote Data Collection & Real-Time Analysis

- Developed an instrumentation box that hosted a **Data Acquisition (DAQ) system** that captured over 900 daily set of acceleration, strain and image data under different environmental condition over 15 months & designed a **digital signal filter** to retrieve true signal from the collected data for mechanical analysis of railway behavior.

WORK EXPERIENCE

SEECO Consultants, Inc **Tinley Park, IL**
Staff Engineer Jan 2020 – Aug 2020

- Designed a total of 8 structural bridges & foundations as well as environmental infrastructure such as wastewater management tank & prepared 12 technical engineering reports as the primary consulting engineer

TECHNICAL SKILLS

Programming: Python, R, Julia, Shell (Bash), SQL, C++, CUDA, Tableau.

REWARDS

Zachry Department of Civil & Environmental Engineering Excellence Fellowship Award, 2021-2024.
Hagler Institute for Advanced Study Heep Graduate Fellowship Award, 2025-2026.

Terracon Graduate Fellowship Award, 2025-2026.

LEADERSHIP POSITIONS

President, Student Chapter of Society of Underwater Technology (SUT) at Texas A&M University, 2023-2024.
President, Student Chapter of Geotechnical Engineering Student Organization (GESO) at University of Illinois Urbana-Champaign, 2018-2019.

PERSONAL PROJECTS

Bayesian Neural Network for Time-Series Forecasting Spring 2023 - Summer 2023

- Developed a **Bayesian Neural Network** in Python using the **LSTM architecture** and the Multiple-Try Metropolis to predict air pollutants concentration with its associated uncertainty and achieved similar prediction as the **Gradient-Descent based optimization**

Bayesian Network vs. Neural Network Regression for COVID-19 Risk Assessment Fall 2022

- Formulated two competing COVID-19 cases prediction approaches and constructed two different risk map based on model simulation results